ECEN 104 – Lab 4: Measuring Electric Fields

Group I

Date : 10/16/25

1. Introduction

The purpose of this experiment was to experimentally characterize and visualize electric field distributions in different conductor geometries using resistive paper. This setup simulates practical engineering applications such as high-voltage transmission systems in the Z-machine, where avoiding dielectric breakdown is crucial. The experiment measured electric potentials across three different geometries—parallel plates, coaxial (circular), and wedge-shaped conductors—and derived local electric field intensities from spatial voltage gradients. The central question addressed whether the electric field anywhere in the wedge configurations would exceed the breakdown threshold of 500 kV/m when scaled to a 20 kV system. Our results found that it would.

2. Theory and Background

The electric field E is related to the spatial gradient of electric potential V by the relation:

$$E = -\nabla V \quad [1]$$

For discrete measurements, the field components can be approximated using finite differences:

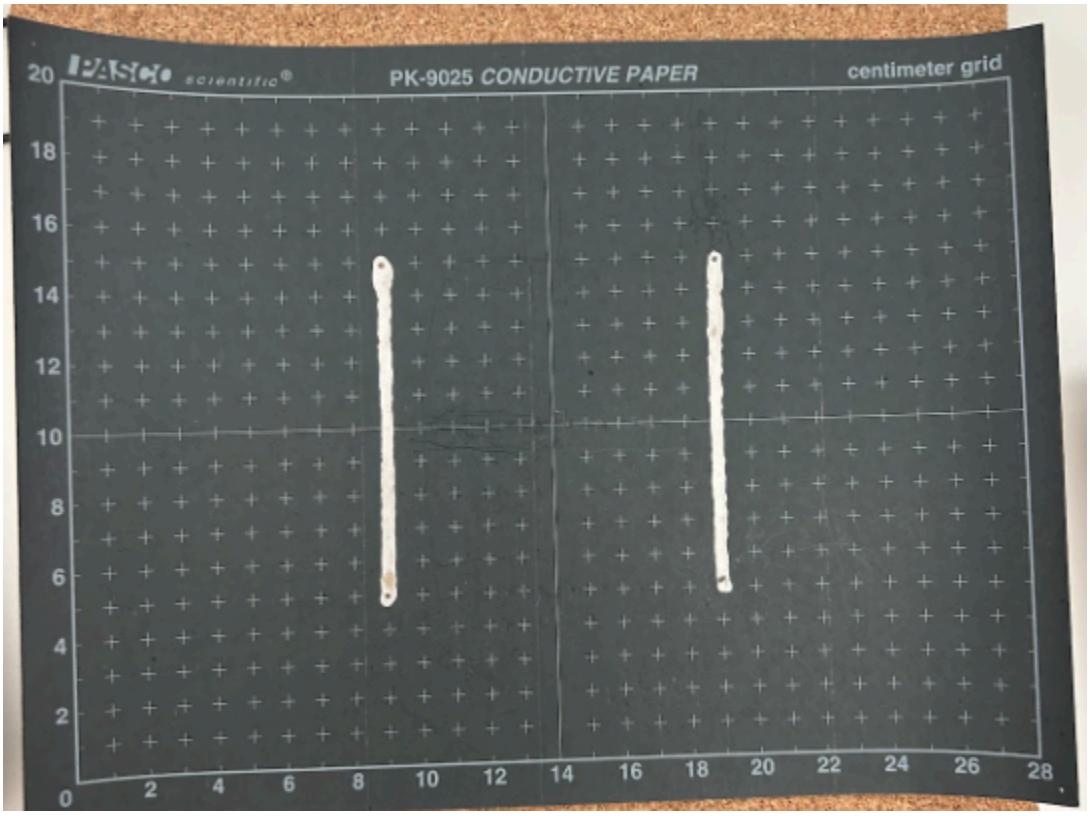
$$E_x \approx -\Delta V/\Delta x \quad [2], \qquad E_y \approx -\Delta V/\Delta y \quad [3]$$

In this experiment, the resistive paper provided a two-dimensional analog to conductive media. Applying voltage across the paper established a potential distribution equivalent to that of infinitely long conductors extending perpendicular to the paper's surface. The potential measurements at various points allowed for the construction of potential contour maps and calculation of corresponding electric field magnitudes.

3. Experimental Procedure

The experiment used resistive paper with silver epoxy conductors and a 5 V power supply. The outer boundary was held at 0 V (ground), while the inner conductor was held at +5 V. Voltage measurements were taken at grid points using a digital multimeter.

Part 1 – Parallel Plate Configuration:



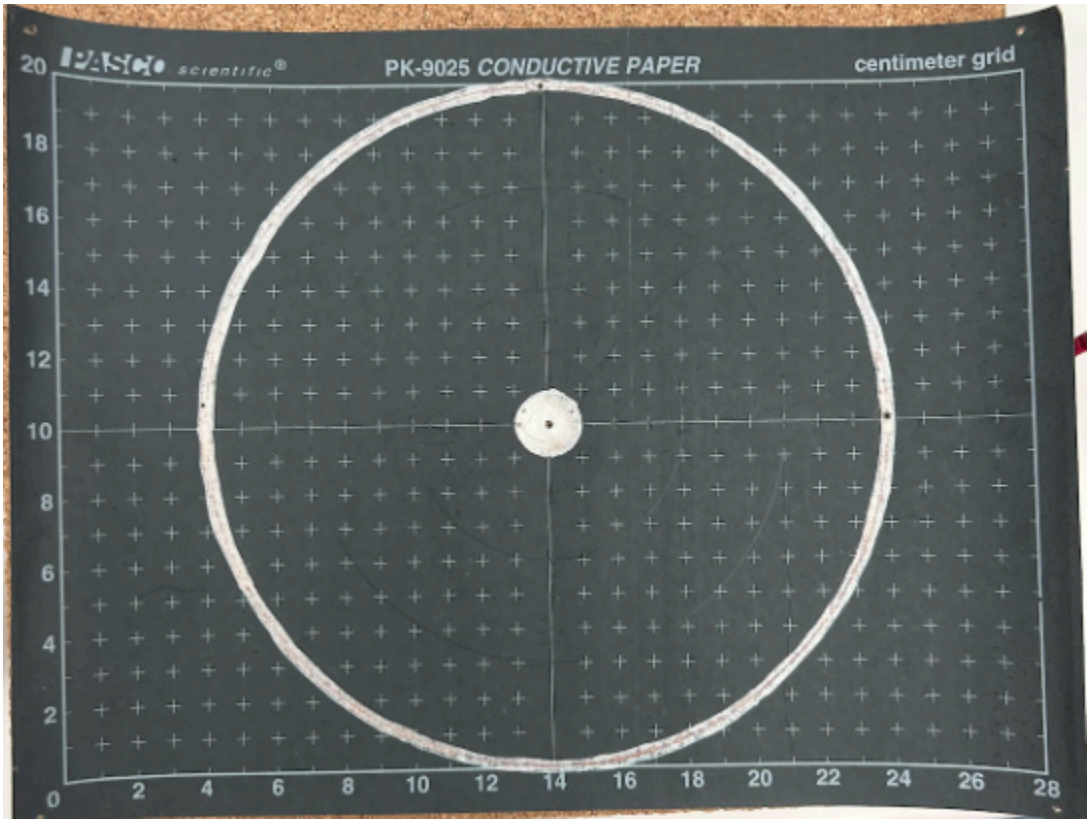
Two rectangular electrodes were drawn at opposite ends of the resistive paper to form a uniform field region. Voltage readings were recorded along four horizontal lines separated by 4 cm vertically and 1cm spacing horizontally.

Vertical is 4 cm separation (4,8,12,16) | Horizontal is 1cm separation (7-21)

4.17	4.24	4.21	3.90	3.43	3.00	2.87	2.15	1.71	1.39	1.16	0.89	0.50	0.42	0.41
4.47	4.57	5.00	4.60	3.56	3.15	2.89	1.90	1.55	1.25	0.86	0.41	0	0.04	0.11
4.35	4.44	4.99	4.51	3.69	3.13	2.84	1.94	1.54	1.14	0.81	0.34	0	0.06	0.11
4.26	4.28	4.34	4.00	3.55	3.25	2.88	1.80	1.61	1.31	1.00	0.66	0.43	0.30	0.33

Table 1: Measured Voltages – Parallel Plate Configuration

Part 2 – Coaxial (Circular) Configuration:



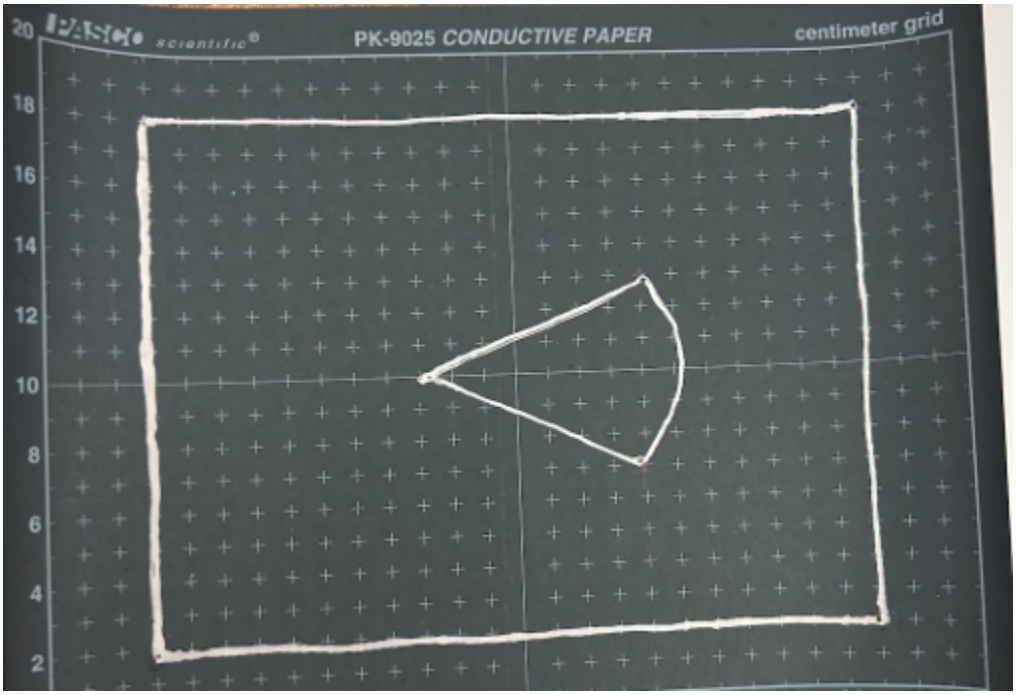
Concentric circular electrodes were used to simulate coaxial geometry. Measurements were taken along two radii (radially upward and downward) at 1 cm spacing to verify symmetry. Only one radius vector needed to be measured because of the nature of circular electric fields, but we did two to show that they are mostly symmetrical.

All 1cm separation radially

Radially Up	5.00	5.00	2.86	2.19	1.63	1.36	1.03	0.70	0.41	0.17	0
Radially Down	5.00	5.00	3.00	2.30	1.60	1.35	0.97	0.81	0.35	0.19	0

Table 2: Measured Voltages – Circular Configuration

Part 3 – Wedge Configuration:



A wedge-shaped inner conductor was drawn inside a rectangular outer conductor to simulate a tapered coaxial system. Voltage readings were taken at multiple horizontal and vertical positions.

Vertical is 4 cm separation (6,10,14) | Horizontal is 1cm separation (7-21)

.076	1.07	1.24	1.54	1.90	2.15	2.37	2.80	3.03	3.09	3.17	3.02	2.77	2.32	1.87
1.13	1.57	2.06	2.52	5.00	4.70	4.70	4.88	4.78	4.80	4.83	4.86	4.84	3.99	2.96
0.77	1.00	1.28	1.51	1.89	2.12	2.38	2.71	3.07	3.25	3.39	2.93	2.94	2.36	1.81

Table 3: Measured Voltages – Wedge Configuration

4. Results

4.1 Parallel Plate Field Distribution

Measured voltages exhibited a near-linear decrease with distance from the positive electrode, confirming a uniform field. Scaling to a 20 kV potential difference across a 4 cm separation yielded a field of approximately 290290 V/m.

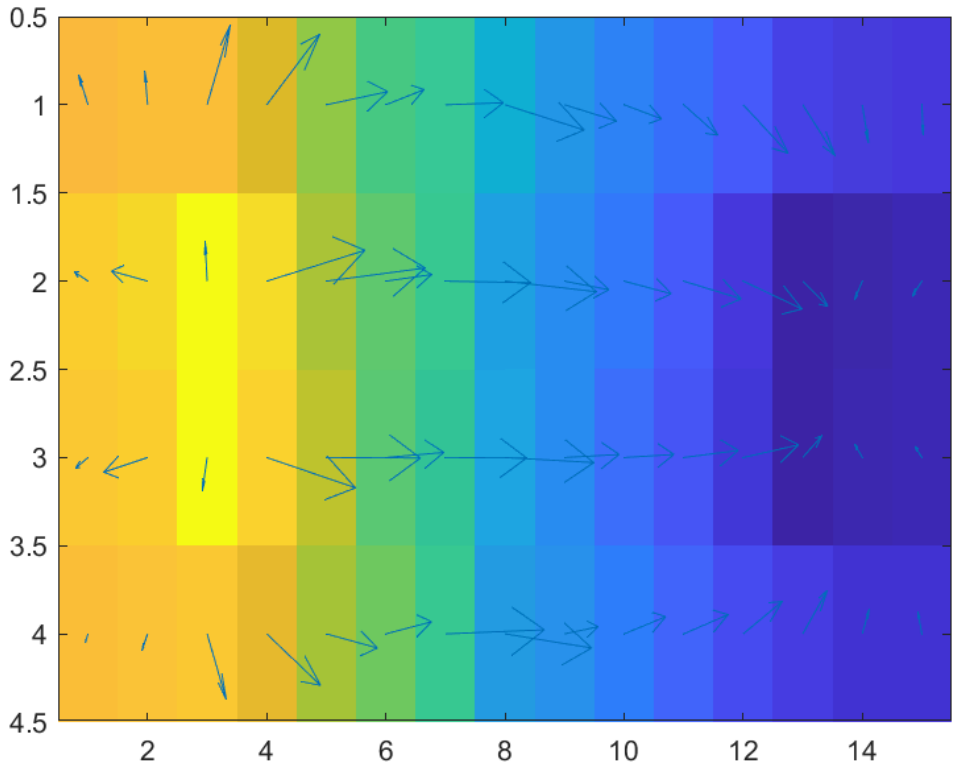


Figure 1: MATLAB Gradient of Parallel Plate Electric Field

4.2 Coaxial (Circular) Geometry

The potential decreased logarithmically with radial distance, consistent with theory for coaxial systems. The scaled maximum electric field strength was 588,000 V/m, slightly exceeding the 500 kV/m breakdown threshold. The highest field occurred near the inner conductor.

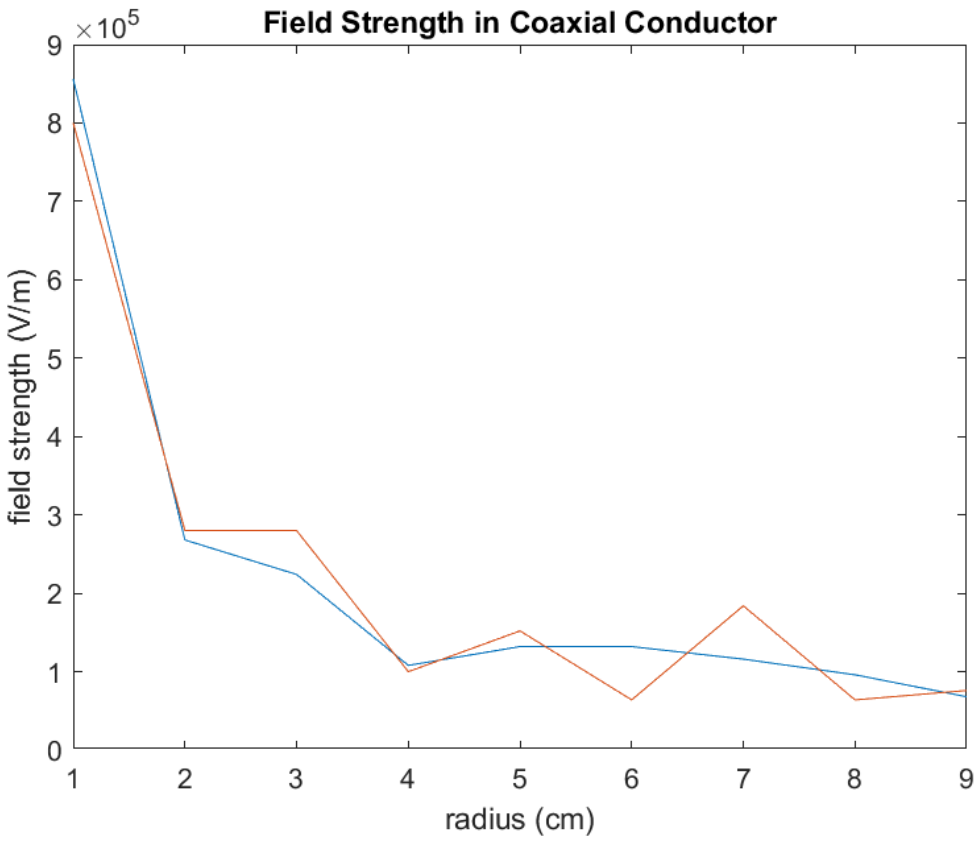


Figure 2: MATLAB Graph Radial Electric Field

4.3 Wedge Geometry

The wedge configuration exhibited especially strong field concentration near the tip.

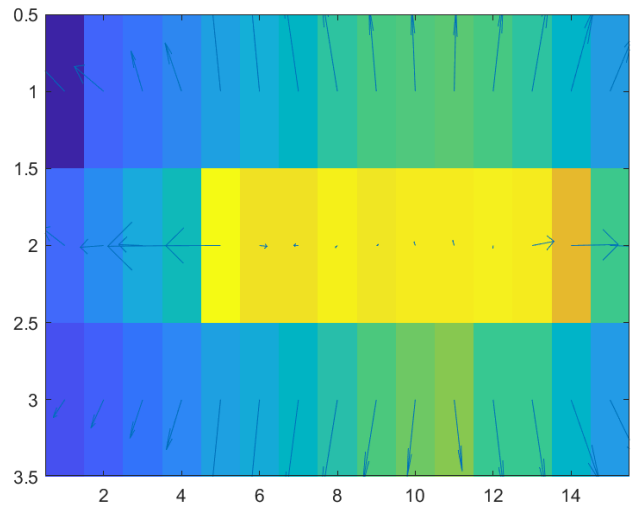


Figure 3: MATLAB Gradient of Wedge Electric Field

We also traced the geometry of the wedge conductor using Photoshop and analyzed the problem using an iterative electric field solver that one of the group members had previously written, which yielded a predicted maximum field of 1.58 MV/m.

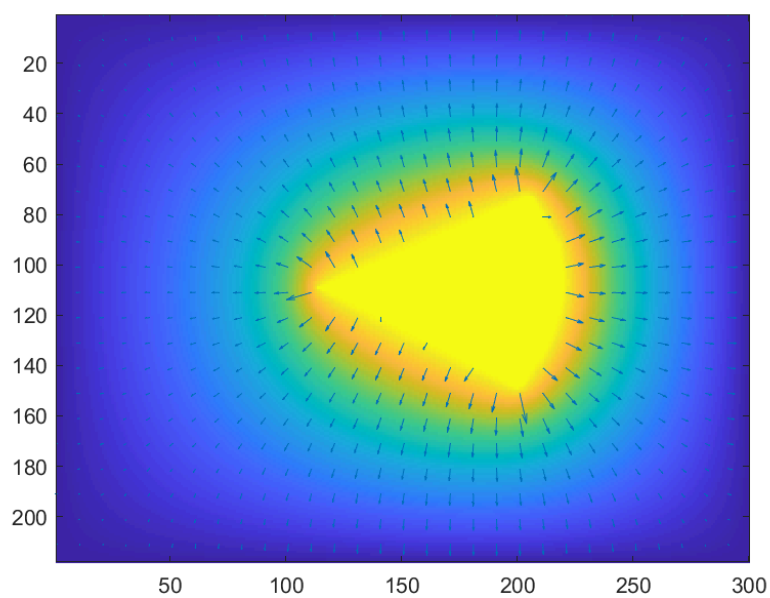


Figure 4: Gradient of MATLAB Iterative + Photoshop E-Field Solver

5. Discussion

The parallel plate experiment validated the linear voltage-field relationship, showing excellent agreement with theoretical predictions. Deviations were attributed to minor edge effects and probe contact resistance. For the coaxial setup, the logarithmic voltage decay and symmetric results along both radii confirmed expected electrostatic behavior. At 20 kV scaling, fields near the inner boundary surpassed the 500 kV/m breakdown limit. In the wedge geometry, sharp corners intensified the electric field dramatically—a phenomenon known as field enhancement. This emphasizes the importance of avoiding acute geometric features in high-voltage design.

6. Conclusions

The resistive paper method proved to be an effective and intuitive way to visualize electric field distributions across multiple conductor geometries. The parallel plate configuration produced a uniform electric field consistent with theoretical predictions, validating the experimental setup and measurement process. In contrast, the coaxial configuration displayed a logarithmic potential variation, with field magnitudes peaking near the inner conductor where the curvature intensified the field concentration. When scaled to a 20 kV potential difference, this region exceeded the dielectric breakdown threshold of 500 kV/m.

The wedge geometry exhibited the most pronounced field enhancement, with measured and simulated maximum intensities reaching approximately 1.6 MV/m. This result highlights how sharp geometric features can greatly amplify local electric fields, presenting significant challenges in high-voltage system design. Overall, the experiment has indicated that XYZ Corp should not proceed with this design. Future design improvements could involve using less sharp edges to minimize electric field crowding.